Boolean Simplification & Duality Principls

1. The Absorption Law

Theory:
The Absorption Law is a powerful shortcut in Boolean algebra. It states that if you have a variable added to the product of its *complement* and another variable, the complement gets "absorbed" (or disappears).

The Rules:

1. $x + \bar{x}y = x + y$
2. $\bar{x} + xy = \bar{x} + y$

---

2. Simplification Practice Problems

Theory:
When simplifying complex expressions, the goal is to rearrange terms so you can factor out common variables, allowing you to use the $A + \bar{A} = 1$ rule to eliminate variables entirely.

Problem 1: Basic Factoring

Equation:

$$
\begin{aligned}
f(x,y) &= x\bar{y} + \bar{x}y + xy + \bar{x}\bar{y} \\
&= \bar{y}(x + \bar{x}) + y(\bar{x} + x) && \text{(Rearrange and factor out } \bar{y} \text{ and } y) \\
&= \bar{y}(1) + y(1) && \text{(Apply Rule: } x + \bar{x} = 1) \\
&= \bar{y} + y \\
&= 1 && \text{(Apply Rule: } y + \bar{y} = 1)
\end{aligned}
$$

Problem 2: Practice

Equation:

$$
\begin{aligned}
&A[B + C(AB + AC)] \\
&= A[B + ABC + ACC] && \text{(Distribute C)} \\
&= A[B + ABC + AC] && \text{(Apply Rule: } C \cdot C = C) \\
&= A[B(1 + AC) + AC] && \text{(Factor out B from first two terms)} \\
&= A[B(1) + AC] && \text{(Apply Rule: } 1 + AC = 1) \\
&= AB + AAC && \text{(Distribute A from the outside)} \\
&= AB + AC && \text{(Apply Rule: } A \cdot A = A) \\
&= A(B + C) && \text{(Factor out A for final answer)}
\end{aligned}
$$

Problem 3: Applying the Absorption Law

Equation:

$$
\begin{aligned}
&\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC \\
&= BC(\bar{A} + A) + \bar{B}\bar{C}(A + \bar{A}) + A\bar{B}C && \text{(Group terms and factor out } BC \text{ and } \bar{B}\bar{C}) \\
&= BC(1) + \bar{B}\bar{C}(1) + A\bar{B}C && \text{(Apply Rule: } A + \bar{A} = 1) \\
&= BC + \bar{B}\bar{C} + A\bar{B}C \\
&= BC + \bar{B}(\bar{C} + AC) && \text{(Factor out } \bar{B} \text{ from the last two terms)} \\
&= BC + \bar{B}(\bar{C} + A) && \text{(Apply Absorption Law: } \bar{C} + AC = \bar{C} + A) \\
&= BC + \bar{B}\bar{C} + A\bar{B} && \text{(Distribute } \bar{B} \text{ back in)}
\end{aligned}
$$

---

3. The Duality Principle

Theory:
The Duality Principle states that if you have a valid Boolean expression or equation, it will remain valid if you convert it to its "dual."

The Conversion Rules:

1. Change every **AND** ($\cdot$) to an **OR** ($+$).
2. Change every **OR** ($+$) to an **AND** ($\cdot$).
3. Change every **1** to a **0**.
4. Change every **0** to a **1**.
   (Important: Variables and their complements do NOT change. A stays A, and $\bar{A}$ stays $\bar{A}$.)

(Proving Dual Identities):
Let's take a standard Boolean rule and prove its dual using the steps you wrote down:

$$
\begin{aligned}
\text{Original Rule:} \quad & A + AB = A \\
\text{Its Dual:} \quad & A \cdot (A + B) = A
\end{aligned}
$$

$$
\begin{aligned}
\text{Proof of Dual:} \quad & A(A + B) \\
&= AA + AB && \text{(Distribute)} \\
&= A + AB && \text{(Rule: } AA = A) \\
&= A && \text{(Rule: } A + AB = A)
\end{aligned}
$$
